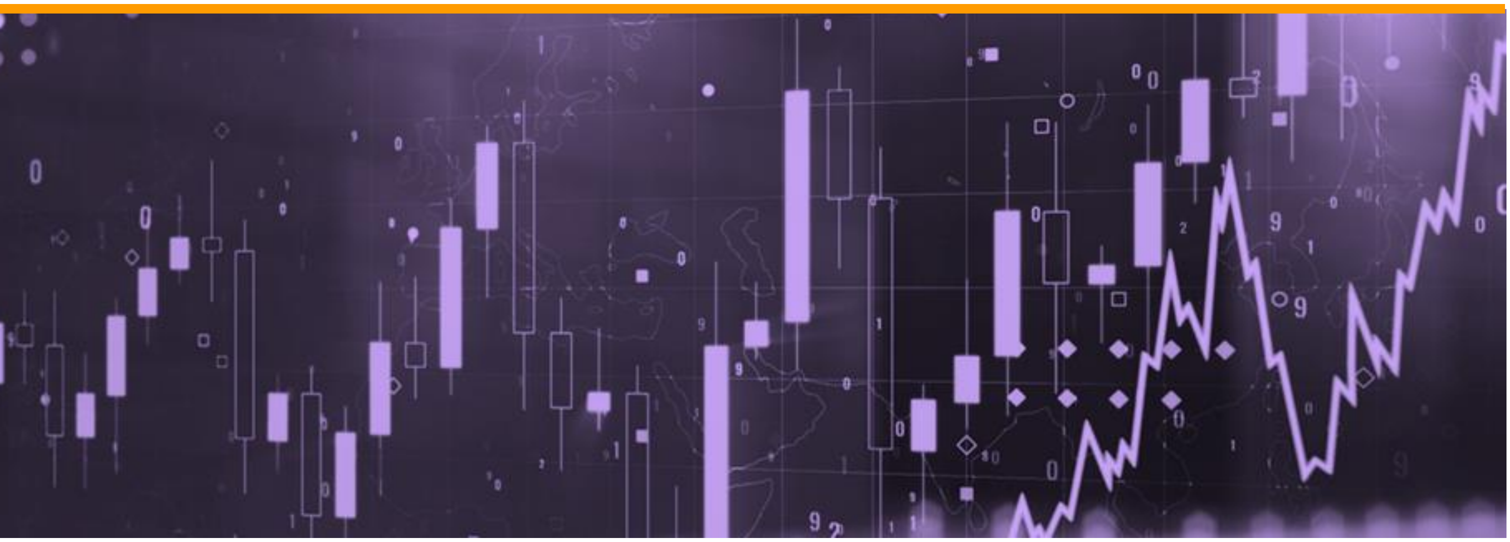

BPI’s Stress Testing Testimony and 2024 Stress Test Results

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On June 26, 2024, BPI testified at a House Financial Services Committee subcommittee hearing to address concerns about the lack of transparency in the Federal Reserve's stress tests and its impact on bank credit availability and cost. Later that day, the Federal Reserve released the 2024 stress test results, revealing higher capital depletions under stress scenarios, which will lead to increased capital requirements for many banks effective Oct. 1, 2024. A key factor contributing to higher capital requirements is the lower projections of net revenues as determined by supervisory models.

During the hearing, BPI presented several examples highlighting the inaccuracies and volatility inherent in the supervisory stress-testing models used to determine the stress capital buffer and the need for increased transparency and notice and comment on the supervisory models. Anticipating the 2024 stress test results, BPI emphasized that one of the main drivers of excess volatility in capital requirements is the lack of granularity in the Fed's revenue projections.

A significant contributor to the excessive variability in stress test projections is the Fed's reliance on aggregated models that assign disproportionate weight to bank performance in the preceding year. These revenue models, based on highly aggregated pre-provision net revenue (PPNR) components, often relegate macroeconomic scenario variables to a secondary role in driving supervisory PPNR projections.[1] As a result, the projections become heavily influenced by backward-looking terms in these models. This "momentum" effect was particularly evident in the Fed's projections of noninterest income and noninterest expense in the 2024 stress test results.

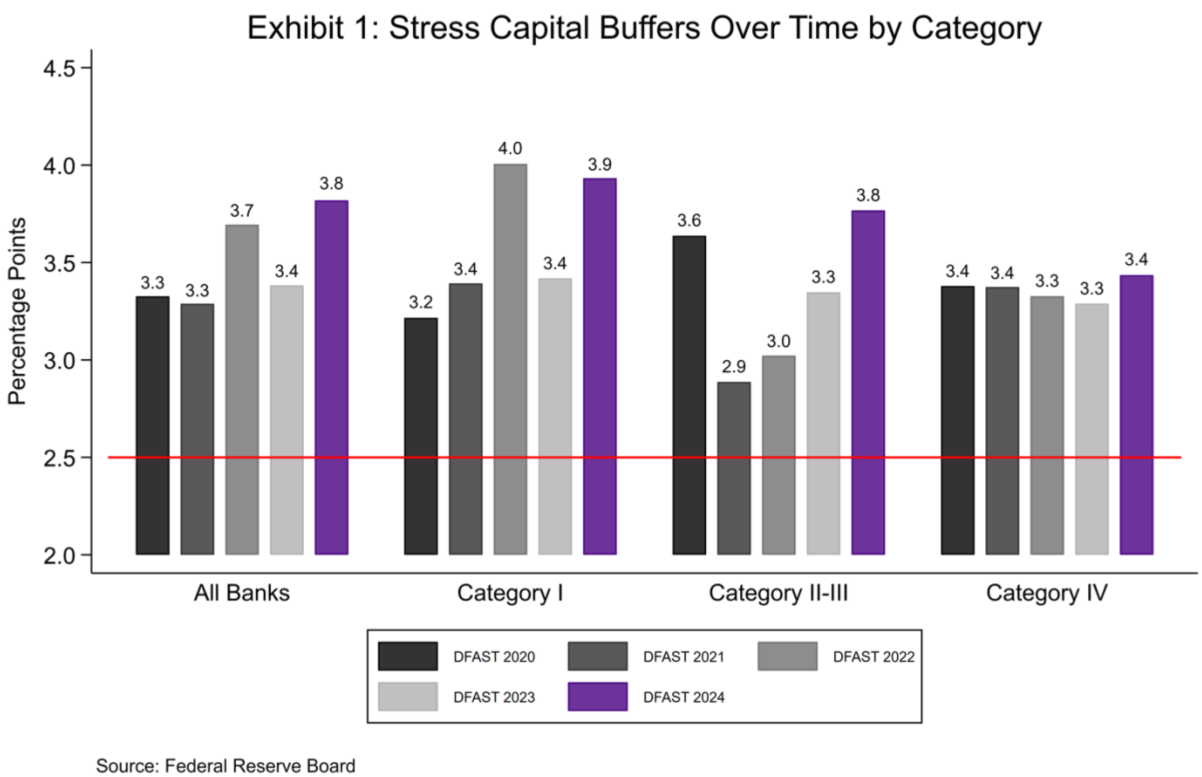
Another noteworthy result from this year’s stress tests is the substantial increase in capital requirements for several intermediate holding companies (IHCs) of foreign banks over the past few years. In 2023, the Fed implemented significant changes to the PPNR modeling for these firms. However, the lack of transparency in the Fed's methodology makes it impossible to understand the factors driving

[1] The role of interest rates in projecting net interest income is an important exception.

the changes in these banks' capital requirements. The main takeaway here is that some of these IHCs have experienced cumulative increases in capital requirements ranging from 6 to 10 percentage points over the past two years, which are extraordinary. These results raise questions about the consistency and fairness of the stress testing process across different types of banking institutions.

Brief Summary of the Results

Based on the 2024 stress test results, we expect a 40-basis-point increase in banks' capital requirements on average (Exhibit 1).[2] The increase varies by bank category, with Category I banks (or the GSIBs) projected to see the largest increase of around 50 basis points. Category II-III banks are expected to see the next largest increase of around 40 basis points, while Category IV banks are looking at an increase of about 10-basis-point increase.



The Federal Reserve's explanation of the stress test results highlights significant changes in banks' financial condition over the past year, accounting for the increase in capital requirements. Two key factors emerged in 2023:

- First, banks experienced a decline in noninterest income coupled with a rise in noninterest expenses. As the performance of financial institutions in the previous year is carried forward in supervisory models regardless of the severely adverse scenario, this year's stress tests reflect lower noninterest income and higher noninterest expenses compared to the previous year's projections.

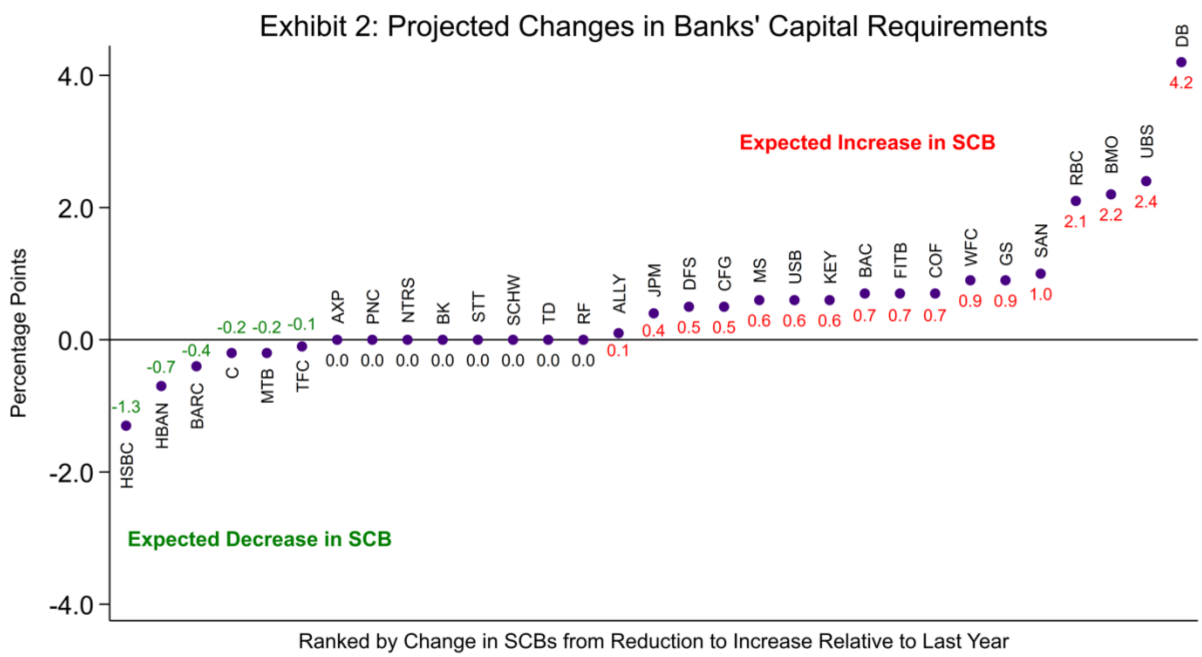
[2] The stress capital buffer projections are based on expectations of dividends, since banks have not yet officially announced their stress capital buffers. The actual capital requirement increases may differ once banks finalize and announce their planned dividends for the following year.

- Second, banks reported a deterioration in loan quality, particularly in commercial and industrial loans and credit card portfolios. This decline in credit quality at the outset of this year's stress tests resulted in higher projected loan losses throughout the stress planning horizon.

Consequently, the combination of reduced projected revenues (due to weaker PPNR) and increased projected losses (stemming from loan quality deterioration) led to greater capital depletion under this year's severely adverse scenario. This will ultimately drive banks' SCBs higher.

Large Variability in Bank Specific Results

Exhibit 2 presents a comprehensive view of changes in bank-specific capital requirements by comparing the stress test outcomes between last year and this year. The banks are ranked from left to right based on their relative performance, with the leftmost bank showing the most improved stress capital buffer compared with the previous test.



Source: Federal Reserve Board, company disclosures, FR Y-9C, BPI calculations.

The results reveal significant variations in SCB changes:

- Over half of the banks face increases in their SCB requirements.
- About one-third of the banks see SCB increases of 70 basis points or more.
- Five banks show SCB increases exceeding 2 percentage points.

To put these findings into perspective, the magnitude of these changes in capital requirements are comparable to the potential impact of the Basel III Endgame proposal from July 2023. That proposal has faced considerable opposition from various stakeholders, including banks, civil rights organizations, farmers, pension funds, small businesses and housing groups.[3] The main difference from the Basel proposal is that, under the SCB framework, banks have only three months to comply with the new

[3] See Latham & Watkins LLP, "Comments on the Basel III Endgame Proposal," February 2, 2024. Available at Comments on the Basel III Endgame Proposal (lw.com)

requirement (as opposed to a much longer Basel III Endgame compliance period) and changes to stress testing scenarios and models are not subject to notice and comment and are undertaken fully at the Fed's discretion.

The substantial fluctuations in capital requirements highlighted by this year's stress test results raise important questions about the consistency and predictability of the stress testing process. These variations could have significant implications for banks' ability to plan and manage their capital effectively, potentially affecting their lending capacity and willingness to participate in trading and market-making activities.

PPNR Projections Are Inaccurate and Volatile

As we noted in our testimony, one important source of the excessive volatility in a bank's SCB is from undue fluctuations in the projections of PPNR, the amount of pre-tax profits a bank earns before deductions taken for expected future loan losses and other losses. These projections are one of the main determinants of bank performance in the stress tests.[4] The modeling of PPNR was an important innovation introduced by the U.S. stress tests, but it also remains a major weakness of the supervisory stress tests.[5] The supervisory stress test models pertaining to PPNR rely heavily on the following criteria:

- Bank-specific effects and autoregressive parameters are overly sensitive to changes in the sample period used to estimate the parameters of the model.
- The macroeconomic variables included in the supervisory scenarios tend to have low explanatory power in most cases (the notable exception being interest rates in the net interest income projections).

The Federal Reserve's description of the PPNR models provides some information about the functional form of the regression models. Based on this description and analyses of prior stress results, we constructed models similar to those used by the Federal Reserve and applied them to generate out-of-sample forecasts. However, as shown in Exhibit 3, because the Fed omits key details in their models, it is impossible to match their projections at the level of granularity required to explain changes in individual banks' capital requirements.

[4] PPNR is defined as net interest income (interest income earned less interest expense) plus noninterest income minus noninterest expense.
[5] See Hirtle, Beverly (2018), "The Past and Future of Supervisory Stress Testing Design." Federal Reserve Bank of New York. Available at https://www.newyorkfed.org/newsevents/speeches/2018/hir181009.

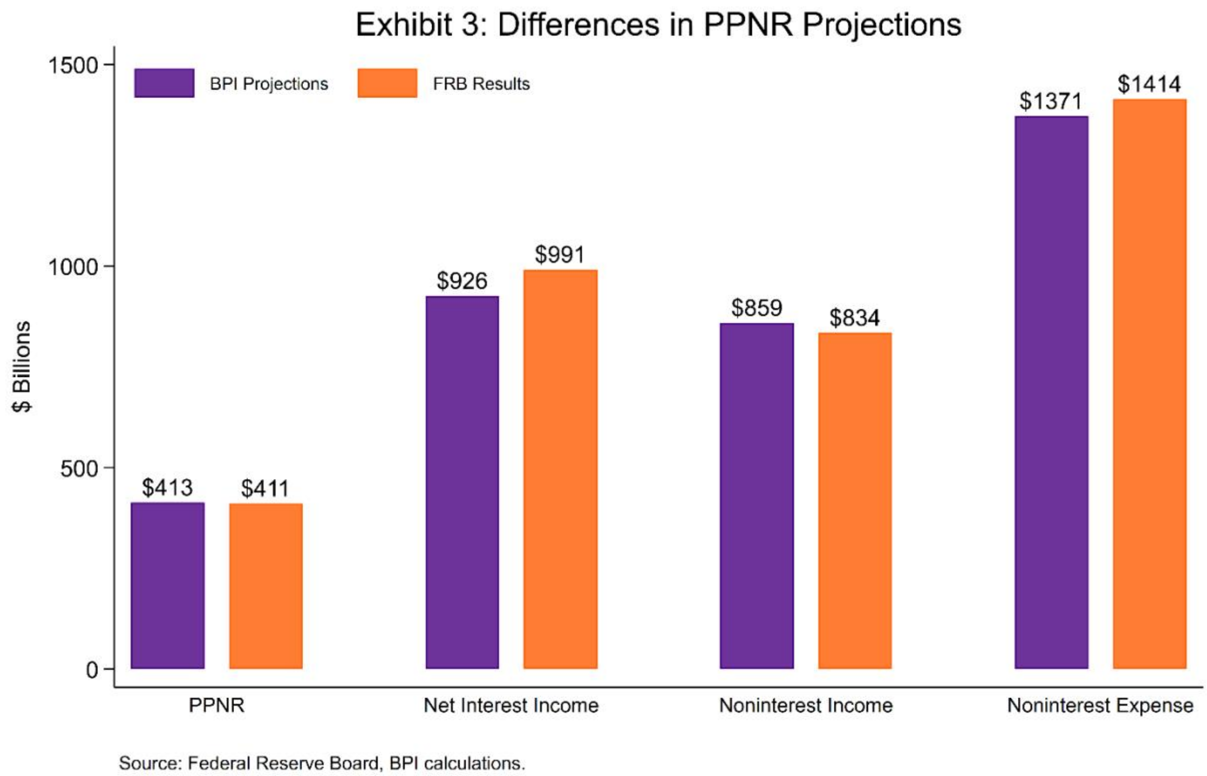


Exhibit 3 compares BPI's projections of the PPNR subcomponents – net interest income, noninterest income, and noninterest expense – against the Fed's results. Although our aggregate PPNR projections are close to the Fed's, we have notable differences across each subcomponent. Specifically, we overestimate noninterest income by $25 billion, underestimate net interest income by $65 billion and project lower noninterest expense by $43 billion.

The main reason for these differences is likely BPI's use of a different sample period in estimating the "bank-specific effects," which aim to capture each firm's average performance in recent years.[6] Without knowing the exact definition of "recent years" used by the Fed, it is impossible to match their projections and, consequently, each bank's individual performance in the stress tests, given that PPNR is among the two or three most important components alongside provisions for loan losses and trading and counterparty losses (with operational risk losses included in PPNR).

There is no strong economic justification for why recent bank performance should play such a disproportionate role in PPNR projections. As we have shown in last year's analysis of stress tests results, banks' own projections for PPNR subcomponents are much more granular and therefore less reliant on recent past performance. As discussed in our testimony, the only way to reduce the excessive volatility in the Fed's projections is to release the supervisory models for notice and comment and allow review by subject matter experts to improve model granularity.

The Fed appears to be taking some steps, albeit modest ones, to improve the granularity of the PPNR models. For example, Box 1 in the stress testing results notes that the Fed is making adjustments to the compensation model that "explicitly condition the projections on the share of variable compensation."

[6] The Fed's stress testing methodology states, "These fixed effects aim to capture individual firm characteristics and differences in business models that cannot be accounted for by firm balance sheet variables."

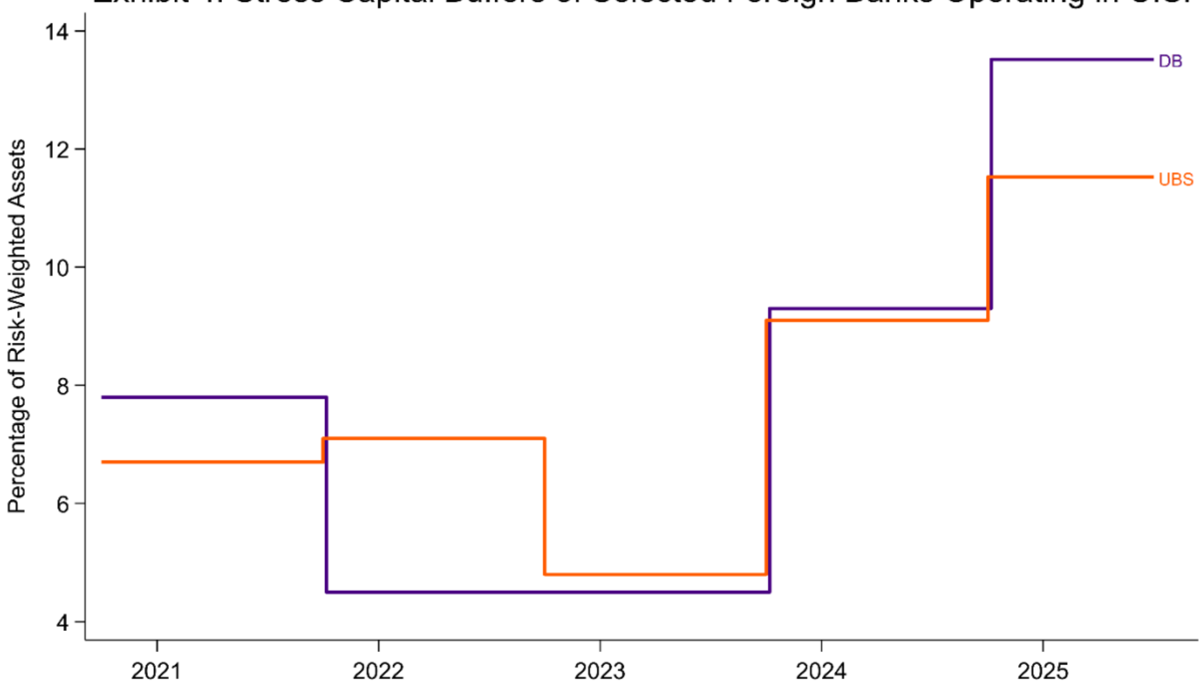
This would reduce the volatility of PPNR projections because variable compensation falls sharply under stress conditions, resulting in lower noninterest expense for banks. However, we do not know the details of the Fed's adjustments, and while they are moving in the right direction, it is unlikely the adjustments go far enough.

Increases in Stress Capital Buffers of IHCs are Extraordinary

Another related issue that appears to be affecting some of the IHCs more dramatically is the Fed’s modelling of PPNR, which results in much higher and more volatile capital charges for these firms. The change in PPNR modeling for IHCs is likely the primary driver of the volatility in the SCBs.

In 2023, the Federal Reserve announced that they had begun individual modeling of PPNR for IHCs. This step was taken because the Fed now had sufficient time series data for these firms to be modeled individually. The new IHCs were formed in 2016, with six firms becoming subject to supervisory stress tests in 2018. From 2018 to 2022, the projections of PPNR for these six IHCs were derived from the industry’s aggregate performance for each revenue and expense component.

Exhibit 4: Stress Capital Buffers of Selected Foreign Banks Operating in U.S.



Source: Federal Reserve Board, Large Bank Capital Requirements, BPI Calculations (2024/2025).

Exhibit 4 illustrates the year-over-year changes in the SCB for two foreign banks operating in the United States. The SCB of DB USA Corporation has increased by nearly 10 percentage points since the Fed announced changes to the PPNR methodology for IHCs. Similarly, the SCB of UBS Americas Holding LLC rose by 7 percentage points over the same period. These are not the only two IHCs experiencing highly unusual increases in capital requirements. As shown in Exhibit 2, the top five increases in capital requirements are all associated with IHCs of foreign banks operating in the U.S.

These substantial increases in capital requirements for IHCs highlight the significant impact of the Fed's methodological changes. *They also raise questions about the consistency and fairness of the stress testing process across different types of banking institutions.*

Summary

In summary, the 2024 stress test results reveal significant challenges in the Federal Reserve's stress testing methodology, particularly in the modeling of PPNR and its impact on banks' capital requirements. The substantial volatility in stress capital buffers, especially for IHCs of foreign banks, raises serious concerns about the consistency, predictability and fairness of the stress testing process.

The Fed's reliance on aggregated models and recent bank performance in PPNR projections, coupled with the lack of transparency in their methodology, makes it difficult for banks to anticipate and plan for changes in capital requirements. Although the Fed appears to be taking some, albeit modest, steps to improve the granularity of their models, these efforts appear insufficient to address the underlying issues.

As noted in BPI's testimony, allowing public comment on scenarios and supervisory models would enable extensive review by experts, academics and banks. This would increase transparency in scenario design and improve model accuracy, fostering a more effective financial system that better serves the U.S. economy's needs.